

Worksheet: Normal Distributions Part III

1. Hobbits are remarkable creatures. The average height of an adult hobbit is 42 inches, and essentially all hobbits are between 30 and 54 inches. If (adult) hobbit heights are normally distributed, what is a good estimate for the standard deviation of this population of heights?
2. Suppose the amount a company's stock price changes each trading day can reasonably be modeled by a normal distribution with a mean $\mu = 0.04$ dollars and a standard deviation of $\sigma = 0.54$ dollars. Estimate the likelihood that on a random trading day, the stock price will increase by more than 1 dollar.
3. Return to the Stock price distribution. Find Q_3 , the value L such that 75% of all trading days will have a change in price less than L .
4. Which is more impressive, a baseball player hitting 30 homeruns in a season in which the league-wide homerun distribution is $N(18, 5)$, or a player hitting 45 homeruns in a season in which the distribution is $N(30, 8)$?

5. Suppose each time I fill up my truck with gas, I record my gas mileage (in miles per gallon) for that tank of gas, and suppose all that data suggests the mileage I get from a tank of gas in my truck varies according to a normal distribution centered at 25 mpg with a standard deviation of 1.5 mpg.
- (a) Using the 68-95-99.7 rule of thumb, estimate the likelihood that the mileage is below 22 mpg.
- (b) Using the 68-95-99.7 rule of thumb, estimate the likelihood that the mileage is between 25 and 28 mpg.
6. A soft-drink machine can be regulated so that it discharges an average of μ oz. per cup. If the ounces of fill are normally distributed, with a standard deviation of 0.4 oz., what value should μ be set at so that 6-oz. cups will overflow only 2% of the time?
7. Find each answer below using the given shaded areas and appealing to the symmetry of a normal distribution. Here X is $N(50, \sigma)$ for some unspecified σ .

(a) $P(X > 40)$

(b) $P(X > 70)$

(c) $P(30 < X < 40)$

(d) $P(X > 70)$

(e) $P(40 < X < 50)$

